RQ Answers

omitted

2024-04-05

Data overview

```
glimpse(all_data)
```

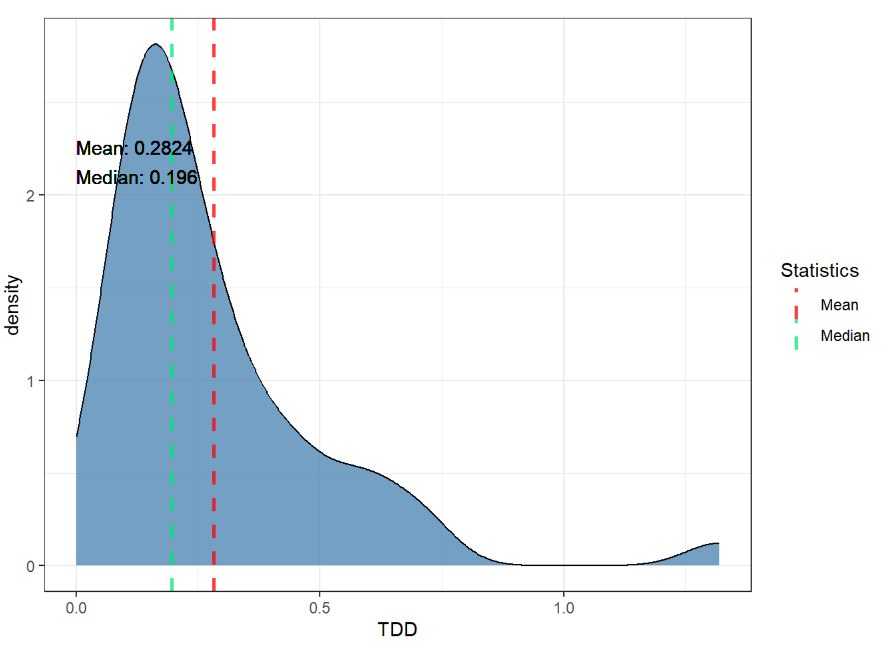
```
## Rows: 900
## Columns: 11
## $ repo              <chr> "cayenne", "cayenne", "cayenne", "cayenne", "cayen…
## $ pr_number         <int> 474, 474, 474, 474, 474, 474, 474, 474, 474, 474, …
## $ rule              <chr> "java:S3776", "java:S3776", "java:S1066", "java:S1…
## $ severity          <chr> "CRITICAL", "CRITICAL", "MAJOR", "MINOR", "MINOR",…
## $ file              <chr> "modeler/cayenne-modeler/src/main/java/org/apache/…
## $ type              <chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM…
## $ status            <chr> "OPEN", "CLOSED", "OPEN", "OPEN", "OPEN", "OPEN", …
## $ debt              <int> 15, 15, 5, 15, 2, 2, 5, 2, 2, 5, 5, 5, 1, 29, 5, 1…
## $ ncloc_affected_file <int> 489, 489, 489, 489, 489, 489, 489, 489, 489, 489, …
## $ complexity        <int> 125, 125, 125, 125, 125, 125, 125, 125, 125, 125, …
## $ origin            <chr> "NEW", "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTI…
```

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the da
ta has 47 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but th
e data has 47 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

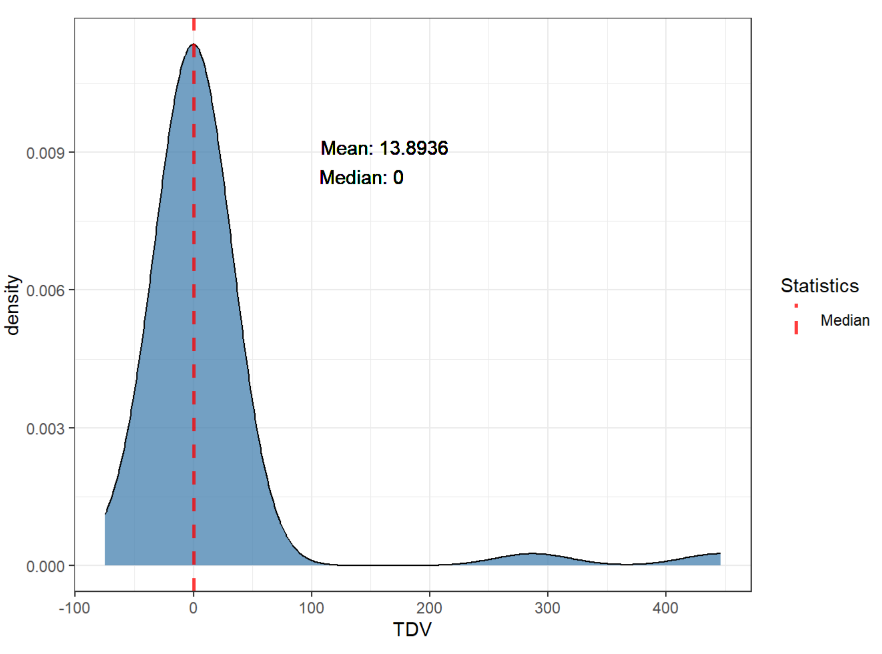


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the da
ta has 47 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but th
e data has 47 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

```
## # A tibble: 1 × 4
##   repo    tdv_lower_than_zero tdv_equal_to_zero tdv_greater_than_zero
##   <chr>                 <dbl>             <dbl>                 <dbl>
## 1 cayenne                12.8              72.3                  14.9
```

Mean of percentages of TDV (mean of repo percentages)

```
## # A tibble: 1 × 3
##   mean_tdv_lower_than_zero mean_tdv_equal_to_zero mean_tdv_greater_than_zero
##                      <dbl>                  <dbl>                      <dbl>
## 1                     12.8                   72.3                       14.9
```

Percentages of pre-existing TDV by repo

```
## # A tibble: 1 × 3
##   repo    preexisting_tdv_equal_to_zero preexisting_tdv_greater_than_zero
##   <chr>                           <dbl>                             <dbl>
## 1 cayenne                          80.8                              19.2
```

Mean of percentages of pre-existing TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   preexisting_tdv_equal_to_zero mean_preexisting_tdv_greater_than_zero
##                           <dbl>                                  <dbl>
## 1                          80.8                                   19.2
```

Percentages for new TDV by repo

```
## # A tibble: 1 × 3
## # Groups:   repo [1]
##   repo    new_tdv_equal_to_zero new_tdv_greater_than_zero
##   <chr>                   <dbl>                     <dbl>
## 1 cayenne                    90                        10
```

Mean of percentages of new TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   mean_new_tdv_equal_to_zero mean_new_tdv_greater_than_zero
##                        <dbl>                          <dbl>
## 1                         90                             10
```

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (../issues_characterization.Rmd), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the $PositionScore$ metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^{k} Position_j$$

given a rule i, its $PositionScore_i$ will be the sum of its position across all k repositiries.After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the PositionScore, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

```
## # A tibble: 10 × 2
## # Groups:   rank [10]
##     rank cayenne
##    <int> <chr>
##  1     1 java:S1121  -  15
##  2     2 java:S1948  -  8
##  3     3 java:S1128  -  3
##  4     4 java:S1135  -  2
##  5     5 java:S1117  -  2
##  6     6 java:S2864  -  1
##  7     7 java:S1192  -  1
##  8     8 java:S3776  -  1
##  9     9 java:S1161  -  1
## 10    10 java:S125  -  1
```

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

```
## # A tibble: 10 × 2
##    rule       position_score
##    <chr>               <dbl>
##  1 java:S1121            122
##  2 java:S1948            123
##  3 java:S1128            124
##  4 java:S1117            125
##  5 java:S1135            126
##  6 java:S106             127
##  7 java:S1161            128
##  8 java:S1192            129
##  9 java:S125             130
## 10 java:S1604            131
```

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

```
## # A tibble: 10 × 2
## # Groups:   rank [10]
##     rank cayenne
##    <int> <chr>
##  1     1 java:S3740  -  73
##  2     2 java:S1948  -  72
##  3     3 java:S3776  -  66
##  4     4 java:S1135  -  55
##  5     5 java:S1066  -  44
##  6     6 java:S1186  -  37
##  7     7 java:S1192  -  34
##  8     8 java:S1133  -  33
##  9     9 java:S1117  -  28
## 10    10 java:S1155  -  27
```

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

```
## # A tibble: 10 × 2
##    rule       position_score
##    <chr>               <dbl>
##  1 java:S3740            122
##  2 java:S1948            123
##  3 java:S3776            124
##  4 java:S1135            125
##  5 java:S1066            126
##  6 java:S1186            127
##  7 java:S1192            128
##  8 java:S1133            129
##  9 java:S1117            130
## 10 java:S1155            131
```

Outliers

Higher TDD

```
## # A tibble: 47 × 5
##    repo    pr_number total_debt total_ncloc   tdd
##    <chr>       <int>      <int>       <int> <dbl>
##  1 cayenne       518        240         182 1.32
##  2 cayenne       531        192         268 0.716
##  3 cayenne       545        316         453 0.698
##  4 cayenne       475        235         389 0.604
##  5 cayenne       516        530         887 0.598
##  6 cayenne       530        107         185 0.578
##  7 cayenne       596        917        1828 0.502
##  8 cayenne       511        397         857 0.463
##  9 cayenne       524        190         421 0.451
## 10 cayenne       522        501        1178 0.425
## # ℹ 37 more rows
```

Higher TDV

```
## # A tibble: 47 × 3
##    pr_number repo      tdv
##        <int> <chr>   <int>
##  1       522 cayenne   446
##  2       536 cayenne   287
##  3       511 cayenne    25
##  4       578 cayenne    10
##  5       553 cayenne     8
##  6       527 cayenne     2
##  7       541 cayenne     2
##  8       474 cayenne     0
##  9       475 cayenne     0
## 10       481 cayenne     0
## # ℹ 37 more rows
```

Lower TDV

```
## # A tibble: 47 × 3
##    pr_number repo      tdv
##        <int> <chr>   <int>
##  1       537 cayenne   -75
##  2       529 cayenne   -15
##  3       596 cayenne   -15
##  4       497 cayenne   -13
##  5       516 cayenne    -5
##  6       542 cayenne    -4
##  7       474 cayenne     0
##  8       475 cayenne     0
##  9       481 cayenne     0
## 10       484 cayenne     0
## # ℹ 37 more rows
```

Extra

Number of PRs with pre-existing TD

```
##    n
## 1 47
```

Number of java:S1192 issues that affect test code

```
##   n
## 1 0
```

Number of java:S1192 issues that affect not test code

```
##    n
## 1 35
```

Summary NCLOC

```
##      ncloc
##  Min.   :  40.0
##  1st Qu.: 193.0
##  Median : 389.0
##  Mean   : 684.2
##  3rd Qu.: 769.5
##  Max.   :4577.0
```